

NAME : _____

CLASS : _____



JURONG PIONEER JUNIOR COLLEGE
JC2 Preliminary Examination 2022

BIOLOGY
Higher 2

9744/03
21 September 2022

Paper 3 Structured and Free-response Questions

2 hours

Candidates answer on the Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your class and name in the spaces at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
Section B	
Total	

This document consists of **17** printed pages.

Section A

Answer **all** the questions in this section.

- 1 Colorectal cancer and hepatocellular carcinoma (HCC) are amongst the leading causes of cancer-related deaths worldwide. Colorectal cancer is the cancer of the colon or rectum. It is amongst the top three causes of cancer in both men and women globally, with more than a million people diagnosed with colorectal cancer each year.

Most colorectal cancers are due to old age and lifestyle factors, with only a small number of cases due to underlying genetic factors.

- (a) State two environmental causes of cancer.

..... [2]

1. Ionising radiation such as nuclear radiation/X-rays/gamma rays/UV radiation ;
2. Chemical carcinogens such as tobacco/tar in cigarette smoke / ethidium bromide / asbestos ;

- (b) Colorectal cancer often begins as an adenoma – a benign tumour that develops at the inner lining of the large intestine. Though benign, Fig. 1.1 shows how adenomas may transform to become malignant as a result of genetic alterations over time.

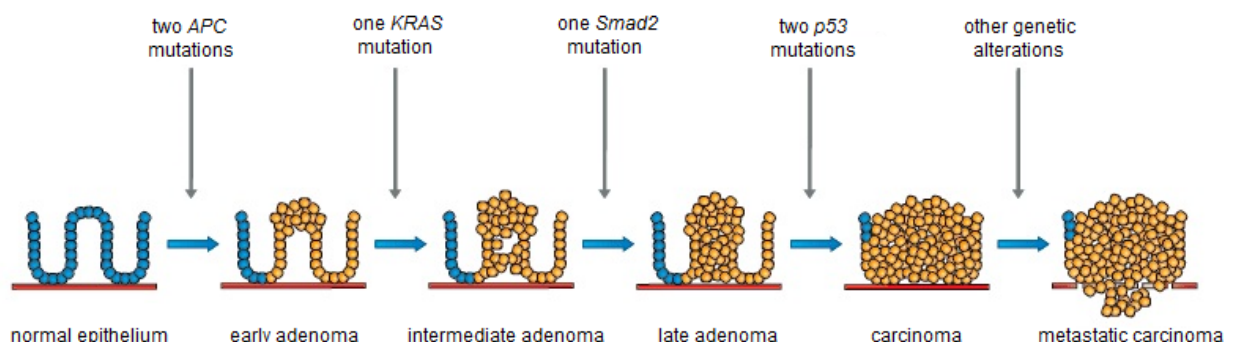


Fig. 1.1

Explain how Fig. 1.1 provides evidence for the development of cancer as a multi-step process.

..... [4]

1. Fig. 1.1 shows that a single cell needs to accumulate more than one (somatic) mutation to become a cancerous cell ;
2. Mutations in the tumour suppressor genes APC / p53 / proto-oncogenes KRAS / Smad2, as well as other genetic alterations in the same somatic cell ;
3. results in the rate of cell division to be greater than the rate of cell death, hence leading to uncontrolled cell division (and tumour formation) ;
4. When cells of a tumour (*accept: carcinoma*) invade surrounding tissue, and metastasise to other sites, the tumour is considered to be malignant and cancer has developed ;

MP1/MP2: reference to single/same somatic cell at least once

- (c) Different types of gene mutation have been found to affect tumour suppressor genes.

Explain why a base pair deletion in a tumour suppressor gene is more likely to produce a non-functional protein than a base pair substitution.

..... [3]

1. Base pair deletion results in a frameshift mutation where the reading frame is altered and nucleotides downstream from the deletion are improperly grouped into (incorrect) codons ;
2. Resulting in extensive missense mutation / primary structure / many amino acids in the polypeptide chain will be changed, which will greatly alter the 3D conformation of the protein ;
3. Whereas for base pair substitution, it may result in a silent mutation where the same amino acid is still coded for due to the degeneracy of the genetic code ;
OR
4. Base pair substitution may result in a missense mutation / a different amino acid is coded for, where the amino acid has similar R-group properties to the original amino acid coded for ;
OR
5. Base pair substitution may result in a missense mutation / a different amino acid is coded for, but the amino acid is not found in a critical region of the protein ;

Pt 1 + Pt 2 + Pt 3/4/5 for full credit

Unlike colorectal cancer, the incidence of HCC can be attributed to different genetic and epigenetic alterations. Epigenetics refers to heritable states of gene expression without alteration to the DNA sequence itself. Epigenetic changes such as DNA hypermethylation or hypomethylation, dysregulation of histone modification patterns, chromatin remodelling and aberrant expression of micro-RNAs (miRNAs) are associated with HCC.

miRNAs are non-coding single-stranded RNAs that are typically 19 to 25 nucleotides in length. They mostly function as negative gene regulators by binding to their complementary sequence located at the 3' untranslated region of their target mRNA, thereby forming miRNA-mRNA duplexes and silencing gene expression.

- (d) State the level at which miRNAs regulate gene expression.

..... [1]

1. Translational level ;

- (e) Fig. 1.2 shows how chronic liver damage and the replacement of normal liver tissue by fibrous scar tissue can lead to liver cirrhosis. Cirrhosis is generally irreversible once it occurs; more often than not, it leads to HCC.

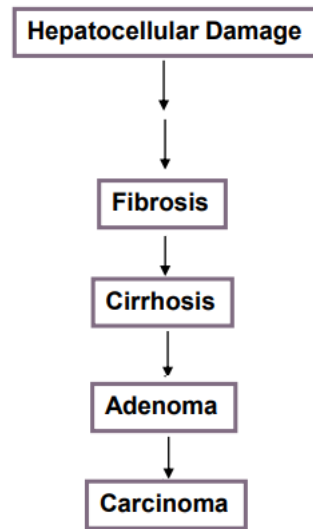


Fig. 1.2

A study investigated the association between differential levels of miRNA expression in liver cirrhosis. Non-cirrhotic and cirrhotic liver samples were used and the results are shown in Fig. 1.3.

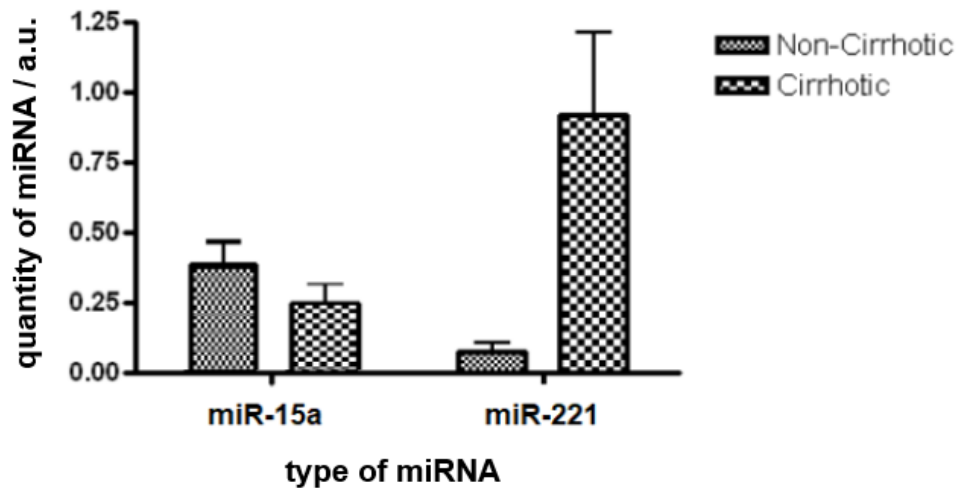


Fig. 1.3

- (i) Compare the effect of the relative quantities of miRNA in leading to liver cirrhosis.

..... [3]

Describe (Quote data)**miR-15a**

1. The quantity of miR-15a decreased from (an average of) 0.40 a.u. in non-cirrhotic liver samples to (an average of) 0.25 a.u. in cirrhotic liver samples / decreased by 0.33% (OTWWE) ;

miR-221

2. The quantity of miR-221 increased from (an average of) 0.07 a.u. in non-cirrhotic liver samples to (an average of) 0.91 a.u. in cirrhotic liver samples / increased by 13 times (OTWWE) ;

Similarity

3. Changes in the relative quantities of each miRNA contributed to the progression towards liver cirrhosis ;

Difference

4. An increase in the quantity of miR-221 and a decrease in the quantity of miR-15a led to liver cirrhosis ;
- (ii) 'oncomiRs' are miRNAs that promote oncogenesis by negatively regulating important tumour suppressor genes. On the other hand, 'anti-oncomiRs' are miRNAs that exert tumour-suppressive effects by repressing oncogenes.

Deduce if miR-15a and miR-221 is an 'oncomiR' or 'anti-oncomiR' and justify your answer.

miR-15a

justification

.....

miR-221

justification

.....

[4]

1. miR-15a: anti-oncomiR ;
2. (*mark for idea*) Down-regulation of miR-15a led to a reduced repression of oncogenes / increased levels of oncogene expression, which was correlated with the progression towards liver cirrhosis ;
3. miR-221: oncomiR ;
4. (*mark for idea*) Up-regulation of miR-221 led to a greater extent of repression of tumour suppressor genes / reduced levels of TS gene expression, which was correlated with the progression towards liver cirrhosis ;

- (f) The molecular genetics of HCC have recently been extensively characterised. One of the most consistent epigenetic changes in HCC is that of DNA hypermethylation at the promoters of target genes.

- (i) Describe how DNA hypermethylation at promoters can alter expression levels of the target genes.

..... [3]

1. DNA (hyper)methylation physically blocks the binding of general transcription factors and RNA polymerase to the promoter (of the target genes) ;
2. hence preventing formation of the transcription initiation complex, preventing transcription ;
3. DNA (hyper)methylation also leads to recruitment of transcriptional repressors / histone deacetylases / other repressive chromatin remodelling complexes, thereby forming more compact chromatin / tighter nucleosomes / DNA more tightly coiled around nucleosomes, preventing transcription ;

MP2/MP3: reference to preventing T/C must be made at least once for full credit

- (ii) The *p16INK4a* gene is located on chromosome 9 and is one of the most frequently altered genes observed in HCC. The prevalence of hypermethylation of its promoter in various types of liver tissues are shown in Tables 1.1 and 1.2.

Table 1.1

sample	percentage of samples which showed <i>p16INK4a</i> promoter hypermethylation
HCC	54.5
non-tumourous	15.6

Table 1.2

sample	percentage of samples which showed <i>p16INK4a</i> promoter hypermethylation
cirrhotic	22.7
non-cirrhotic	9.2

Predict whether *p16INK4a* is an oncogene or tumour suppressor gene and suggest a possible role of its protein product in the cell cycle.

..... [2]

1. ***p16INK4A* is a tumour suppressor gene ;**

2. **Role of protein product – inhibit cell cycle progression / triggers cell cycle arrest (OWTTE) ;**
3. **AVP – triggers apoptosis etc. ;**

(g) Another common hallmark of HCC is that of telomerase reactivation.

A student made the following claim:

“In eukaryotes, telomerase synthesises additional lengths of DNA that are added to the ends of chromosomes, thus solving the end-replication problem that results from each round of DNA replication.”

(i) Account for the end-replication problem.

..... [4]

1. **DNA polymerase can only add DNA nucleotides to the free 3' –OH end of an existing strand ;**
2. **With the removal of the RNA primer from the 5' end of the newly synthesised strand ;**
3. **There is no free 3' (–OH) end for DNA polymerase to add (free) nucleotides to / the RNA primer is not replaced (with DNA nucleotides) ;**
4. **This results in a daughter strand that is shorter than the parental/template DNA strand / progressive shortening of telomeres (in eukaryotes) / a 3' overhang at the end of the chromosome ;**

(ii) Evaluate the validity of the student's claim that the addition of DNA to chromosomal ends solves the end-replication problem.

..... [2]

Not valid

1. **The end-replication problem can never be solved / shortening of chromosomal ends in (linear) eukaryotic chromosomes continues to occur with each round of DNA replication due to the removal of the RNA primer (OWTTE) ;**

Valid to a certain extent in that:

2. **With each round of DNA replication, the shortening of telomeres occurs without deleterious effects / genes within the chromosome will not be lost / loss of important genetic information is prevented (OWTTE) ;**

(iii) Telomerase is **not** present in prokaryotic cells. Suggest why prokaryotes do **not** have telomerase.

..... [2]

1. **Prokaryotes have circular DNA / do not have linear chromosomes / do not have telomeres (A: DNA has no ends) ;**
2. **Thus they do not encounter the end-replication problem ; so there is no need for telomerase in prokaryotic cells.**

[Total: 30]

- 2 Gene expression in eukaryotes is regulated at different levels and involves various regulatory proteins and enzymes. At translational level of regulation, regulatory proteins such as translation initiation factors and translational repressors are involved.

(a) Describe how gene expression may be regulated by translational repressors.

..... [2]

1. Translational repressors recognise and bind to specific sequences, usually the 5' untranslated region (5' UTR) of mature mRNA ;
2. Blocking / preventing ribosomes from binding to mature mRNA so that initiation of translation cannot take place ;

Another regulatory protein, ubiquitin, is a highly conserved protein comprising 76 amino acids, ubiquitously expressed in all tissues in eukaryotes.

(b) State the role of ubiquitin in regulating gene expression in eukaryotes.

..... [1]

1. Marks proteins for degradation by proteasome ;

Fig. 2.1 shows the structure of an ubiquitin molecule.

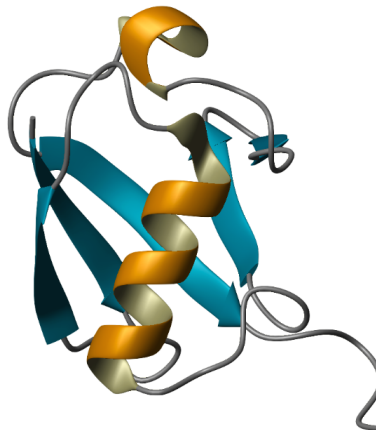


Fig 2.1

(c) Describe **two** similarities between the secondary structures found in ubiquitin.

..... [2]

Both α -helices and β -pleated sheets

1. are maintained by (intramolecular) hydrogen bonds between the (O of) C=O group and (the H of the) -NH group in the polypeptide backbone ;
 2. contain hydrogen bonds formed at regular intervals in the polypeptide chain ;
- OR
3. are geometrically regular repeating structures in the polypeptide chain ;
 4. These hydrogen bonds are disrupted by heat / changes in pH ;

α -helices and β -pleated sheets must be mentioned for full credit

Ubiquitin is synthesised at cytosolic ribosomes. mRNA, tRNA and rRNA are involved in the synthesis of ubiquitin.

(d) Describe the roles of tRNA in the synthesis of ubiquitin.

..... [2]

1. To act as an intermediate molecule between the codon of mRNA and the amino acid sequence of the polypeptide chain ;
2. To carry the correct amino acid from the cytoplasm to the polypeptide chain being synthesised at the ribosome ;

During protein synthesis, transcription precedes translation. Researchers measured the error rates during transcription and translation in yeast cells. The results are shown in Table 2.1.

Table 2.1

process	error rate
transcription	10^{-5} per base
translation	10^{-3} per codon

(e) Suggest why such a difference in error rates during transcription and translation is tolerated by the yeast cell.

..... [3]

1. Error rate during transcription is lower than error rate during translation (or vice versa) ;
2. Each error in transcription leads to the subsequent translation of the mRNA molecule with an incorrect sequence into many copies of non-functional protein ;
3. Whereas each error during translation only results in one copy of non-functional protein ;

[Total: 10]

- 3 Algae are aquatic photosynthetic protists. Some researchers genetically modified the unicellular alga, *Chlorella vulgaris*, to try to increase the rate of the light independent stage of photosynthesis.

C. vulgaris was modified to increase the expression of the gene coding for aldolase. Aldolase is an enzyme that causes an increase in the concentration of rubisco.

Two cultures of *C. vulgaris*, one that was not genetically modified (unmodified) and one genetically modified, were grown under controlled conditions for 14 days. Samples were taken from the cultures at regular intervals during the 14 days to obtain measurements of dry mass. The results are shown in Fig. 3.1.

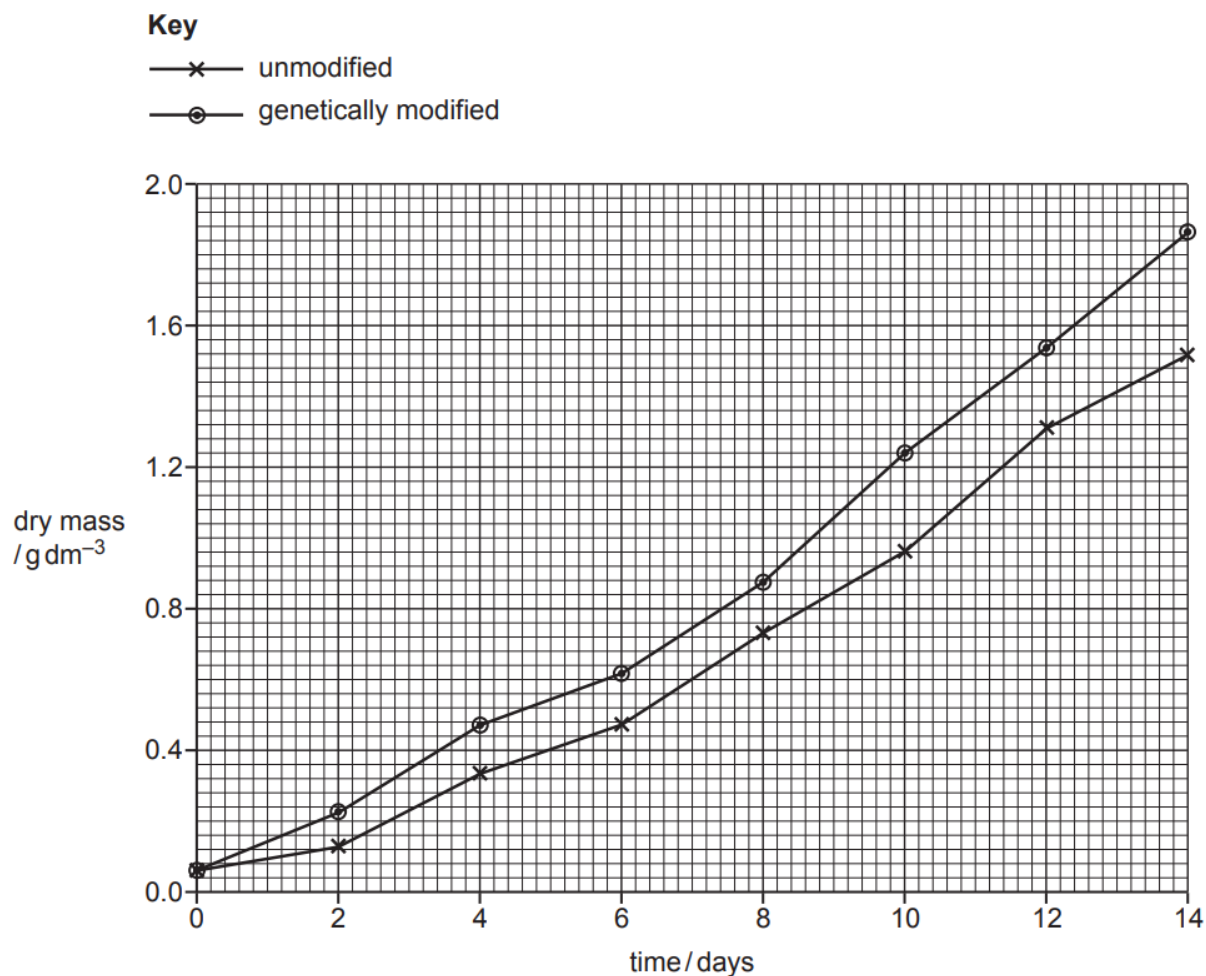


Fig. 3.1

- (a) With reference to Fig. 3.1, describe the differences between the results for the two cultures.

..... [3]

1. genetically modified (GM) culture has higher (dry) mass throughout the experiment than unmodified culture + quote relevant data from day 2 to 14
 - e.g. At day 2, dry mass of GM culture is 0.24 g dm^{-3} , higher than the dry mass of unmodified culture of 0.12 g dm^{-3} ;
2. GM culture has a (slightly) higher rate of/steeper, increase in dry mass ;
3. the largest difference in (dry) mass between the 2 culture is after day 8 ;
4. processed data to show difference (see table for computed values) ;
5. AVP: rate of growth over 14 days
 - rate of growth for GM $\rightarrow 1.86 - 0.08 / 14 = 0.13 \text{ g dm}^{-3} \text{ day}^{-1}$
 - rate of growth for unmodified $\rightarrow 1.52 - 0.08 / 14 = 0.10 \text{ g dm}^{-3} \text{ day}^{-1}$

time / days	dry mass / g dm^{-3}		
	genetically modified ± 0.02	unmodified ± 0.02	difference between the two ± 0.04
2	0.22	0.13	0.09
4	0.48	0.34	0.14
6	0.62	0.48	0.14
8	0.88	0.74	0.14
10	1.24	0.96	0.28
12	1.54	1.31	0.23
14	1.86	1.52	0.34

- (b) Explain how the Calvin cycle was affected by the genetic modification of *C. vulgaris*.

..... [3]

1. More rubisco leads to greater rate of/more carbon dioxide fixation ;
2. Greater rate of glycerate phosphate (GP) / PGA produced ;
3. Greater rate of triose phosphate produced (from GP/ PGA);
4. Greater rate of/more regeneration of RuBP;
5. Greater rate of/more Calvin cycle ;

- (c) Intermediate products of the Calvin cycle are needed to produce organic molecules for use by the cell. One such organic molecule is phospholipid, a major constituent of cell membranes.

Outline the functions of membranes within cells.

..... [3]

1. **Membranes of cell organelles enable separate compartments to be formed inside the cell, thereby allowing specialised metabolic pathways to take place ;**
 2. **Selective barrier – the hydrophobic core of membranes acts as a selective barrier, restricting the movement of charged ions or polar molecules across the membrane ;**
 3. **Transport – channel proteins or carrier proteins embedded in membranes regulate the movement of polar molecules/charged ions into and out of the cell/organelles ;**
 4. **Signal transduction (cell signalling) – specific signals on the extracellular surface of the cell are detected and relayed to the inside of the cell, triggering a specific cellular response within the cell ;**
 5. **Membranes are the site where enzymes may be embedded and ordered to carry out sequential steps of a metabolic pathway ;**
 6. **Membranes allow for transport proteins/enzymes/receptors to be embedded, leading to an increase in surface area for metabolic reactions to occur efficiently ;**
- (d) Planting large numbers of trees is one way to reduce global atmospheric CO₂ concentration. Large scale culture of genetically modified *C. vulgaris* could also reduce global atmospheric CO₂ concentration.

Suggest one advantage of using genetically modified *C. vulgaris* instead of trees to reduce global atmospheric CO₂ concentration.

..... [1]

1. **Faster growth rate / reproduction ;**
2. **Cheaper to set up ;**
3. **Can culture algae in lab / not using land to grow algae / not constrained by lack of space ;**

[Total: 10]

NAME : _____

CLASS : _____

Section BAnswer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in parts **(a)** and **(b)**, as indicated in the question.

- 4 (a) Describe how transformation, transduction and conjugation give rise to variation in prokaryotic genomes and explain why genetic variation is important in bacteria. [15]

Transformation

1. A (competent) bacterium takes up foreign DNA ;
2. The foreign DNA is incorporated into bacterium's chromosome via homologous recombination ;
3. By crossing over where there is sufficient homology between the DNA fragments and bacterial chromosome ;
4. Segments of the bacterium's genome are replaced ;

Transduction

5. Bacterial DNA / genes are transferred from one bacterium to another via a bacteriophage ;
6. When a (virulent) bacteriophage undergoes lytic cycle, a small piece of the host bacterial cell's degraded DNA is (mistakenly) packaged within the capsid of a defective phage ;
7. Defective phage infects another bacterial cell and injects the piece of host bacterial DNA into the newly infected bacterial cell cytoplasm ;
8. When a (temperate) bacteriophage enters into lytic cycle from lysogenic cycle, a small region of the host bacterial DNA that was adjacent to the prophage is improperly excised and the phage-host hybrid DNA is packaged within the capsid (of a defective phage);
9. The defective phage infects another bacterial cell and injects the phage-host hybrid DNA into the newly infected bacterial cell cytoplasm ;
10. The host bacterial DNA/ phage-host hybrid DNA is incorporated into recipient bacterium's genome / DNA via homologous recombination ;

Conjugation

11. F⁺ donor cell produces sex pilus to attach to F⁻ recipient cell ;
12. Sex pilus retracts upon contact, pulling the two cells closer and forming a temporary cytoplasmic mating bridge between F⁺ donor cell and F⁻ recipient cell ;
13. A single strand of F plasmid breaks at a specific point (origin of transfer) and the F plasmid is transferred as a single strand ;
14. From F⁺ donor cell into the F⁻ recipient cell via the cytoplasmic mating bridge ;
15. Each single stranded DNA in each respective cell (donor and recipient cells) now acts as a template for the synthesis of a complementary daughter strand ;
16. F⁻ recipient cell (without F plasmid) is now a F⁺ cell ;

Why genetic variation is important

17. Genetic variation is important in bacteria so that bacteria may acquire new alleles / genes ;
18. That confer selective advantage under a particular selection pressure (e.g. antibiotic resistance is a selective advantage, in the presence of antibiotics) ;
19. Bacteria with selective advantage are selected for and reproduce (via binary fission) to pass on favourable alleles to subsequent generations ;

- Pt 1-16: max 13
- Pt 1-19: max 14

QWC [1]

Clearly expressed and well structured, with all 4 parts of question addressed using correct terminology.

- (b) Describe the structures and organisation of viral, prokaryotic and eukaryotic genomes. [10]

Viral genome

1. A virus contains nucleic acid either DNA or RNA as its genome but never both ;
2. Viral genome might consists of a single nucleic acid molecule or several nucleic acid molecules (i.e. segmented) ;
3. The nucleic acid molecules may be circular or linear ;
4. The nucleic acid can be single-stranded (ss) or double-stranded (ds) ;
5. The genetic material may sometimes be associated directly with nucleoproteins, known as nucleocapsid ;
6. Genes ranges from 4 to 100 base pairs (depending on the type of virus) ;

Prokaryotic genome

7. Genome is usually found on 1 main chromosome (excluding plasmids) ;
8. Circular / Looped chromosome ;
9. The chromosomal DNA must be compacted by forming looped domain and supercoiling of the loop DNA ;
10. Naked / associated with nucleoid-associated proteins (H-NS proteins) ;
11. Small genome size / fewer genes present compared to eukaryotic genome ;
12. Presence of operons / cluster of genes controlled by single promoter ;

Eukaryotic genome

13. Genome is divided into many different chromosomes (no. of chromosomes is species-dependent) ;
14. Linear chromosomes ;
15. Large genome size / more genes present ;
16. (Very high degree of compaction) “beads-on-a-string” where DNA winds around histone proteins to form nucleosomes. Nucleosomes undergo further packing to form 30 nm chromatin fibre → 300 nm chromatin fibre.
17. May further condense during nuclear division ;
18. DNA associates with histones (forming nucleosomes) / scaffolding proteins ;
19. Presence of introns interspersed within a gene ;
20. Presence of centromeres / telomeres in a linear chromosome;
21. Besides promoters, eukaryotes typically contain enhancers and/or silencers as regulatory sequence in their DNA ;
22. Genes involved in the same pathway are usually separated on different chromosomes, with each gene under the control of 1 promoter ;

- MP 1-6: max 3
- MP 7-12: max 3
- MP 13-21: max 3

QWC [1]

Clearly expressed and well structured to address structures of all 3 genomes and organisation of prokaryotic & eukaryotic genomes using correct terminology.

[Total: 25]

- 5 (a) Outline and contrast chemiosmosis in photosynthesis and respiration. [15]

During photosynthesis and respiration in aerobic conditions,

1. Flow of electrons down the electron transport chain (ETC) releases energy to pump H^+ across the membrane (via active transport) ;
2. generating a proton gradient across the membrane ;
3. ref. membrane impermeable to H^+ ;
4. Chemiosmosis occurs when H^+ diffuse down their (electrochemical and) proton gradient ;
5. through the ATP synthase ;
6. Releasing (electrical potential) energy ;
7. which drives the phosphorylation of ADP to ATP ;
8. catalysed by ATP synthase ;

Differences between chemiosmosis in photosynthesis and respiration:

Features		Photosynthesis		Respiration
occurs during	9.	photophosphorylation ;	10.	oxidative phosphorylation ;
location	11.	occurs at thylakoid membrane of chloroplast ;	12.	occurs at the inner mitochondrial membrane ;
source of energy for ATP synthesis	13.	energy for ATP synthesis comes from light ;	14.	energy for ATP synthesis comes from glucose oxidation ;
source of electrons	15.	from water	16.	from reduced NAD and reduced FAD
accumulation of protons in	17.	thylakoid lumen / space ;	18.	intermembrane space ;
flow of protons through ATP synthase	19.	from thylakoid lumen to stroma ;	20.	from intermembrane space mitochondrial matrix ;

- Pt 1-20: max 14

QWC [1]

Clearly expressed, using correct terminology to address chemiosmosis and marking clear features of contrast between chemiosmosis in photosynthesis and respiration.

- (b) Describe how the molecular structure and properties of cellulose are related to its functions. [10]

Structure

1. A cellulose molecule consists of a (long) straight chain ;
2. of (many) β -glucose residues ;
3. joined by β -1,4 glycosidic bonds ;
4. Successive β -glucose residues are rotated 180° with respect to the adjacent residue ;
5. which results in the -OH groups projecting outwards from each cellulose chain in all directions ;
6. Hydrogen bonds are formed between the -OH groups of (neighbouring) cellulose chains ;
7. lying in parallel ;
8. resulting in (extensive) cross-linking ;
9. that binds the chains rigidly together to form microfibrils and macrofibrils ;

Properties

10. which confer high tensile strength, stability and support ;
11. AVP – (the extensive hydrogen bonds) result in cellulose being insoluble (in water) ;

Function

12. Cellulose is the main structural component of plant cell walls ;
13. The high tensile strength of microfibrils prevents the plant cell from bursting/lysing when water enters by osmosis ;
14. As a plant cell inflates with water, pressure develops inside it and the cell becomes turgid (turgid cells help support plants which lack wood) ;
15. The arrangement of fibres around the cell helps to determine the shape of the plant cell as it grows ;
16. AVP – the arrangement of fibres at angles/criss-cross allows passage of water (A: named substances) / makes cell wall permeable ;

- Pt 1-11: max 9
- Pt 11 & 16: max 1
- Award for accurately drawn + annotated diagram if MP not scored for response in prose

QWC [1]

Clearly expressed using the correct terminology and communicated without ambiguity to describe molecular structure, properties and function of cellulose ;

[Total: 25]